

Machine Learning-Lecture 11

P1.py

```
C:\ml\L11>python p1.py
```

```
  GENDER AGE ... CHEST PAIN LUNG_CANCER
0     M  69 ...      2     YES
1     M  74 ...      2     YES
2     F  59 ...      2     NO
3     M  63 ...      2     NO
4     F  63 ...      1     NO
..  ... ..
304   F  56 ...      1     YES
305   M  70 ...      2     YES
306   M  58 ...      2     YES
307   M  67 ...      2     YES
308   M  62 ...      1     YES
```

```
[309 rows x 16 columns]
```

```
GENDER      0
AGE          0
SMOKING      0
YELLOW_FINGERS  0
ANXIETY      0
PEER_PRESSURE  0
CHRONIC DISEASE  0
FATIGUE      0
ALLERGY      0
WHEEZING     0
ALCOHOL CONSUMING  0
COUGHING     0
SHORTNESS OF BREATH  0
```

SWALLOWING DIFFICULTY 0

CHEST PAIN 0

LUNG_CANCER 0

dtype: int64

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 309 entries, 0 to 308

Data columns (total 16 columns):

#	Column	Non-Null Count	Dtype
0	GENDER	309 non-null	object
1	AGE	309 non-null	int64
2	SMOKING	309 non-null	int64
3	YELLOW_FINGERS	309 non-null	int64
4	ANXIETY	309 non-null	int64
5	PEER_PRESSURE	309 non-null	int64
6	CHRONIC DISEASE	309 non-null	int64
7	FATIGUE	309 non-null	int64
8	ALLERGY	309 non-null	int64
9	WHEEZING	309 non-null	int64
10	ALCOHOL CONSUMING	309 non-null	int64
11	COUGHING	309 non-null	int64
12	SHORTNESS OF BREATH	309 non-null	int64
13	SWALLOWING DIFFICULTY	309 non-null	int64
14	CHEST PAIN	309 non-null	int64
15	LUNG_CANCER	309 non-null	object

dtypes: int64(14), object(2)

memory usage: 38.8+ KB

None

AGE SMOKING ... GENDER_F GENDER_M

0 69 1 ... False True

1 74 2 ... False True

```

2  59    1 ...  True  False
3  63    2 ...  False  True
4  63    1 ...  True  False
.. ...    ... ..
304 56    1 ...  True  False
305 70    2 ...  False  True
306 58    2 ...  False  True
307 67    2 ...  False  True
308 62    1 ...  False  True

```

[309 rows x 16 columns]

```

[[0.72727273 0.    1.    ... 1.    0.    1.    ]
 [0.8030303  1.    0.    ... 1.    0.    1.    ]
 [0.57575758 0.    0.    ... 1.    1.    0.    ]
 ...
 [0.56060606 1.    0.    ... 1.    0.    1.    ]
 [0.6969697  1.    0.    ... 1.    0.    1.    ]
 [0.62121212 0.    0.    ... 0.    0.    1.    ]]

```

LUNG_CANCER

YES 270

NO 39

Name: count, dtype: int64

```

precision recall f1-score support

NO      0.60    0.75    0.67      4
YES     0.99    0.97    0.98     74
accuracy                0.96    78
macro avg    0.79    0.86    0.82    78
weighted avg    0.97    0.96    0.96    78

```

['NO']

P2.py

```
C:\ml\L11>python p2.py
```

```
  GENDER AGE ... CHEST PAIN LUNG_CANCER
0     M  69 ...      2     YES
1     M  74 ...      2     YES
2     F  59 ...      2     NO
3     M  63 ...      2     NO
4     F  63 ...      1     NO
..  ... ..
304    F  56 ...      1     YES
305    M  70 ...      2     YES
306    M  58 ...      2     YES
307    M  67 ...      2     YES
308    M  62 ...      1     YES
```

```
[309 rows x 16 columns]
```

```
  GENDER      0
  AGE        0
  SMOKING     0
  YELLOW_FINGERS  0
  ANXIETY     0
  PEER_PRESSURE  0
  CHRONIC DISEASE  0
  FATIGUE     0
  ALLERGY     0
  WHEEZING    0
  ALCOHOL CONSUMING  0
  COUGHING    0
  SHORTNESS OF BREATH  0
  SWALLOWING DIFFICULTY  0
  CHEST PAIN   0
```

LUNG_CANCER 0

dtype: int64

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 309 entries, 0 to 308

Data columns (total 16 columns):

#	Column	Non-Null Count	Dtype
0	GENDER	309 non-null	object
1	AGE	309 non-null	int64
2	SMOKING	309 non-null	int64
3	YELLOW_FINGERS	309 non-null	int64
4	ANXIETY	309 non-null	int64
5	PEER_PRESSURE	309 non-null	int64
6	CHRONIC DISEASE	309 non-null	int64
7	FATIGUE	309 non-null	int64
8	ALLERGY	309 non-null	int64
9	WHEEZING	309 non-null	int64
10	ALCOHOL CONSUMING	309 non-null	int64
11	COUGHING	309 non-null	int64
12	SHORTNESS OF BREATH	309 non-null	int64
13	SWALLOWING DIFFICULTY	309 non-null	int64
14	CHEST PAIN	309 non-null	int64
15	LUNG_CANCER	309 non-null	object

dtypes: int64(14), object(2)

memory usage: 38.8+ KB

None

['GENDER']

LUNG_CANCER

YES 270

NO 39

Name: count, dtype: int64

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

NO	1.00	0.38	0.55	8
----	------	------	------	---

YES	0.93	1.00	0.97	70
-----	------	------	------	----

accuracy			0.94	78
----------	--	--	------	----

macro avg	0.97	0.69	0.76	78
-----------	------	------	------	----

weighted avg	0.94	0.94	0.92	78
--------------	------	------	------	----

['YES']

P3.py

C:\ml\L11>python p3.py

enter height 5.8

enter age 37

[69.33333333]

(array([[0.31186159, 0.32926944, 0.32926944]]), array([[7, 5, 0]]))